

BIT 1204

# Fundamentals of Website Development

**HTML table basics**

# What is a table?

- A table is a structured set of data made up of rows and columns (**tabular data**).
- A table allows you to quickly and easily look up values that indicate some kind of connection between different types of data, for example a person and their age, or a day of the week, or the timetable for a local swimming pool.

| Person | Age |
|--------|-----|
| Chris  | 38  |
| Dennis | 45  |
| Sarah  | 29  |
| Karen  | 47  |

| Monday                        | Tuesday                                 | Wednesday                                      | Thursday                      | Friday                                         | Saturday                                                   | Sunday                       |
|-------------------------------|-----------------------------------------|------------------------------------------------|-------------------------------|------------------------------------------------|------------------------------------------------------------|------------------------------|
| Public Swim<br>06:30 - 10:30  | Public Swim<br>06:30 - 09:00            | Public Swim<br>06:30 - 09:00                   | Public Swim<br>06:30 - 11:15  | Public Swim<br>06:30 - 09:00                   | Lane Swim<br>08:00 - 09:00                                 | Lane Swim<br>08:00 - 09:00   |
| Aquacise<br>10:30 - 11:15     | Aqua Jog<br>09:15 - 10:00               | Education<br>Swimming Lessons<br>09:00 - 12:00 | Aquacise<br>11:15 - 12:00     | Education<br>Swimming Lessons<br>09:00 - 12:00 | Oldham Active<br>Kids Swimming<br>Lessons<br>09:00 - 13:00 | Public Swim<br>09:00 - 11:00 |
| Lane Swim<br>11:30 - 13:00    | Parent & Baby<br>Class<br>09:30 - 10:15 | Lane Swim<br>12:00 - 13:00                     | Lane Swim<br>12:00 - 13:00    | Lane Swim<br>12:00 - 13:00                     | Parent and Baby<br>12:00 - 12:45                           | Aquacise<br>11:00 - 11:45    |
| Education<br>Swimming Lessons | Public Swim<br>10:00 - 11:45            | Public Swim<br>13:00 - 14:00                   | Education<br>Swimming Lessons | Oldham Active<br>Kids Swimming                 | Public Swim<br>13:00 - 17:00                               | Public Swim<br>11:45 - 13:00 |

# HTML Table

- A table in HTML consists of table cells inside rows and columns.
- The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.
- The HTML tables are created using the **<table>** tag in which the **<tr>** tag is used to create table rows and **<td>** tag is used to create data cells.
- The elements under **<td>** are regular and left aligned by default

# HTML Table Tags

| Tag                     | Description                                                             |
|-------------------------|-------------------------------------------------------------------------|
| <u>&lt;table&gt;</u>    | Defines a table                                                         |
| <u>&lt;th&gt;</u>       | Defines a header cell in a table                                        |
| <u>&lt;tr&gt;</u>       | Defines a row in a table                                                |
| <u>&lt;td&gt;</u>       | Defines a cell in a table                                               |
| <u>&lt;caption&gt;</u>  | Defines a table caption                                                 |
| <u>&lt;colgroup&gt;</u> | Specifies a group of one or more columns in a table for formatting      |
| <u>&lt;col&gt;</u>      | Specifies column properties for each column within a <colgroup> element |
| <u>&lt;thead&gt;</u>    | Groups the header content in a table                                    |
| <u>&lt;tbody&gt;</u>    | Groups the body content in a table                                      |
| <u>&lt;tfoot&gt;</u>    | Groups the footer content in a table                                    |

# Table Cells

Each table cell is defined by a `<td>` and a `</td>` tag.

td stands for table data.

Everything between `<td>` and `</td>` are the content of the table cell.

- **Note:** A table cell can contain all sorts of HTML elements: text, images, lists, links, other tables, etc.

```
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
</table>
```

---

# Table Rows

Each table row starts with a `<tr>` and ends with a `</tr>` tag.  
tr stands for table row.

```
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

---

# Table Headers

Sometimes you want your cells to be table header cells. In those cases use the `<th>` tag instead of the `<td>` tag:

`th` stands for table header.

```
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
    <th>Person 3</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

---

# Border Attribute

- Here, the **border** is an attribute of <table> tag and it is used to put a border across all the cells. If you do not need a border, then you can use border = "0".

```
<body>
```

```
<table border = "1">
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Salary</th>
```

```
</tr>
```

```
</table>
```

# Background Attribute

- An example of using **background** attribute. Here we will use an image available in /images directory.

```
<html>

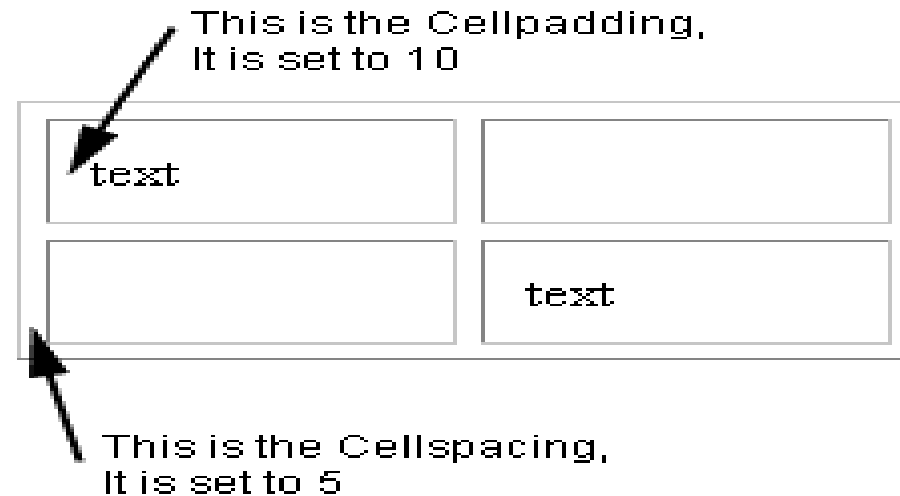
<head>
  <title>HTML Table Background</title>
</head>

<body>
  <table border = "1" bordercolor = "green" background = "/images/test.png">
    <tr>
      <th>Column 1</th>
      <th>Column 2</th>
      <th>Column 3</th>
    </tr>
    <tr>
      <td rowspan = "2">Row 1 Cell 1</td>
      <td>Row 1 Cell 2</td><td>Row 1 Cell 3</td>
    </tr>
    <tr>
      <td>Row 2 Cell 2</td>
      <td>Row 2 Cell 3</td>
    </tr>
    <tr>
      <td colspan = "3">Row 3 Cell 1</td>
    </tr>
  </table>
</body>
```

| Column 1     | Column 2     | Column 3     |
|--------------|--------------|--------------|
| Row 1 Cell 1 | Row 1 Cell 2 | Row 1 Cell 3 |
|              | Row 2 Cell 2 | Row 2 Cell 3 |
| Row 3 Cell 1 |              |              |

# Cellpadding and Cellspacing Attributes

- There are two attributes called *cellpadding* and *cellspacing* which you will use to adjust the white space in your table cells.
- The cellspacing attribute defines space between table cells, while cellpadding represents the distance between cell borders and the content within a cell.



```
<html>

  <head>
    <title>HTML Table Cellpadding</title>
  </head>

  <body>
    <table border = "1" cellpadding = "5" cellspacing = "5">
      <tr>
        <th>Name</th>
        <th>Salary</th>
      </tr>
      <tr>
        <td>Ramesh Raman</td>
        <td>5000</td>
      </tr>
      <tr>
        <td>Shabbir Hussein</td>
        <td>7000</td>
      </tr>
    </table>
  </body>

</html>
```

# Colspan and Rowspan Attributes

- You will use **colspan** attribute if you want to merge two or more columns into a single column. Similar way you will use **rowspan** if you want to merge two or more rows.

|         |  |
|---------|--|
| Colspan |  |
|         |  |

|  |         |
|--|---------|
|  | Rowspan |
|  |         |

Table with column and row spanned:

|           |            |           |             |
|-----------|------------|-----------|-------------|
| Logo      | Top Menu   |           |             |
| Side Menu | World News | Financial | List of Ads |
|           | Technology | Sports    |             |
| Footer    |            |           |             |

```
<head>
  <title>HTML Table Colspan/Rowspan</title>
</head>

<body>
  <table border = "1">
    <tr>
      <th>Column 1</th>
      <th>Column 2</th>
      <th>Column 3</th>
    </tr>
    <tr>
      <td rowspan = "2">Row 1 Cell 1</td>
      <td>Row 1 Cell 2</td>
      <td>Row 1 Cell 3</td>
    </tr>
    <tr>
      <td>Row 2 Cell 2</td>
      <td>Row 2 Cell 3</td>
    </tr>
    <tr>
      <td colspan = "3">Row 3 Cell 1</td>
    </tr>
  </table>
</body>

</html>
```

| Column 1     | Column 2     | Column 3     |
|--------------|--------------|--------------|
| Row 1 Cell 1 | Row 1 Cell 2 | Row 1 Cell 3 |
|              | Row 2 Cell 2 | Row 2 Cell 3 |
| Row 3 Cell 1 |              |              |

# Table Height and Width

- The width and height of a table are defined by the width and height properties.
- You can set a table width and height using **width** and **height** attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

```
<html>

<head>
  <title>HTML Table Width/Height</title>
</head>

<body>
  <table border = "1" width = "400" height = "150">
    <tr>
      <td>Row 1, Column 1</td>
      <td>Row 1, Column 2</td>
    </tr>

    <tr>
      <td>Row 2, Column 1</td>
      <td>Row 2, Column 2</td>
    </tr>
  </table>
</body>

</html>
```

|                 |                 |
|-----------------|-----------------|
| Row 1, Column 1 | Row 1, Column 2 |
| Row 2, Column 1 | Row 2, Column 2 |

# Use the **style** attribute with the **width** or **height**

- Use the style attribute with the width or height properties to specify the size of a table, row or column

```
<table style="width:100%" border="1">
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
  <tr style="height:200px">
    <td>Jill</td>
    <td>Smith</td>
    <td>50</td>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>94</td>
  </tr>
</table>
```

| Firstname | Lastname | Age |
|-----------|----------|-----|
| Jill      | Smith    | 50  |
| Eve       | Jackson  | 94  |

# Nested Tables

- You can use one table inside another table. Not only tables you can use almost all the tags inside table data tag <td>.
- **Example**
- Following is the example of using another table and other tags inside a table cell.

| <table><thead><tr><th>Name</th><th>Salary</th></tr></thead><tbody><tr><td>Ramesh Raman</td><td>5000</td></tr><tr><td>Shabbir Hussein</td><td>7000</td></tr></tbody></table> |        | Name | Salary | Ramesh Raman | 5000 | Shabbir Hussein | 7000 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|--------|--------------|------|-----------------|------|
| Name                                                                                                                                                                        | Salary |      |        |              |      |                 |      |
| Ramesh Raman                                                                                                                                                                | 5000   |      |        |              |      |                 |      |
| Shabbir Hussein                                                                                                                                                             | 7000   |      |        |              |      |                 |      |

# HTML Table Colgroup

- The `<colgroup>` element is used to style specific columns of a table
  - If you want to style the two first columns of a table, use the `<colgroup>` and `<col>` elements
- The `<colgroup>` element should be used as a container for the column specifications. Each group is specified with a `<col>` element.
- The span attribute specifies how many columns that get the style.
- The style attribute specifies the style to give the columns

| MON | TUE | WED | THU | FRI | SAT | SUN |
|-----|-----|-----|-----|-----|-----|-----|
| 1   | 2   | 3   | 4   | 5   | 6   | 7   |
| 8   | 9   | 10  | 11  | 12  | 13  | 14  |
| 15  | 16  | 17  | 18  | 19  | 20  | 21  |
| 22  | 23  | 24  | 25  | 26  | 27  | 28  |

**Note:** The `<colgroup>` tag must be a child of a `<table>` element and should be placed before any other table elements, like `<thead>`, `<tr>`, `<td>` etc., but after the `<caption>` element, if present.

```
<table>
  <colgroup>
    <col span="2" style="background-color: #D6EEEE">
  </colgroup>
  <tr>
    <th>MON</th>
    <th>TUE</th>
    <th>WED</th>
    <th>THU</th>
    ...
```



| MON | TUE | WED | THU | FRI | SAT | SUN |
|-----|-----|-----|-----|-----|-----|-----|
| 1   | 2   | 3   | 4   | 5   | 6   | 7   |
| 8   | 9   | 10  | 11  | 12  | 13  | 14  |
| 15  | 16  | 17  | 18  | 19  | 20  | 21  |
| 22  | 23  | 24  | 25  | 26  | 27  | 28  |

# Nested Tables

```
<body>
  <table border = "1" width = "100%">

    <tr>
      <td>
        <table border = "1" width = "100%">
          <tr>
            <th>Name</th>
            <th>Salary</th>
          </tr>
          <tr>
            <td>Ramesh Raman</td>
            <td>5000</td>
          </tr>
          <tr>
            <td>Shabbir Hussein</td>
            <td>7000</td>
          </tr>
        </table>
      </td>
    </tr>

  </table>
</body>
```